CONFIDENTIAL

DIPLOMA IN INFORMATION TECHNOLOGY

MID-TERM EXAMINATION

COURSE	**:**	**DISCRETE MATHEMATICS**
COURSE CODE	**:**	**DIT 1063**
DATE	**:**	**JAN 26**
Time	**:**	**45 min**

INSTRUCTION TO CANDIDATES

1. This question paper consists of **SEVEN(7)** questions.
2. Answer **ALL** questions.
3. Do not bring any material or electronic devices into the examination hall unless permission is given by the invigilator.

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LECTURER : Mr.Shanmuga Pillai

Do not open the question paper until you are told to do so.

The question paper consists of 2 pages including the cover page.

1. Let P be the statement “It is Sunny” and Q be the statement “I go for a Walk”. Write down in words the following statements, and simplify if possible: **(4 Marks)**

a) $\sim P$

It's not sunny

b) $\sim P \wedge Q$

It’s not sunny and I will go for a walk

c) $P \Rightarrow Q$

If it’s sunny then I will go for a walk

d) $P \wedge Q$

It is sunny and I will go for a walk

2. Write down the truth tables for $(\sim p \vee q) \wedge q$ and $(\sim p \wedge q) \vee q$. **(4 Marks)**

p	**q**	**~p**	**(~p ∨ q)**	**(~p ∧ q)**	**(~p ∨ q) ∧ q**	**(~p ∧ q) ∨ q**
T	T	F	T	F	T	T
T	F	F	F	F	F	F
F	T	T	T	T	T	T
F	F	T	T	F	F	F

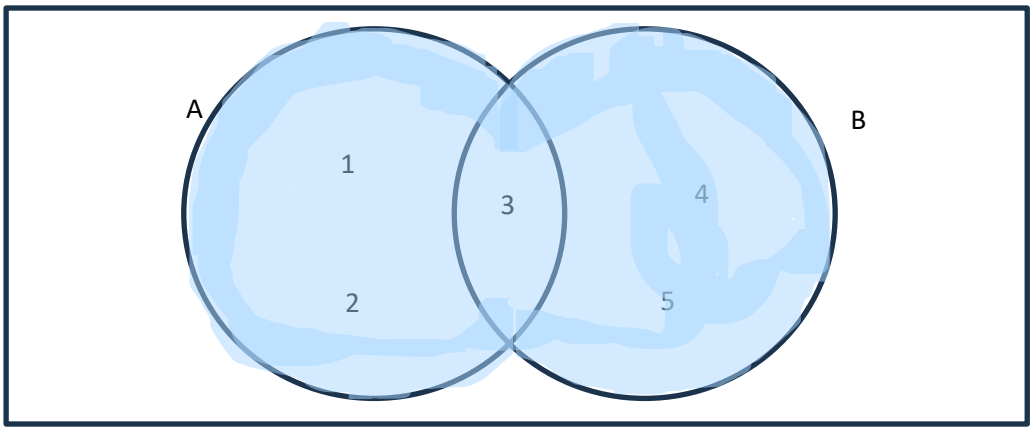
3. In a class of 30 students, 15 students like mathematics, 12 students like science, and 5 students like both subjects. Calculate the probability that a randomly chosen student likes either mathematics or science. **(4 Marks)**

The number of students who like either science or maths= 15+12-5=22

P= 22/30= 11/15

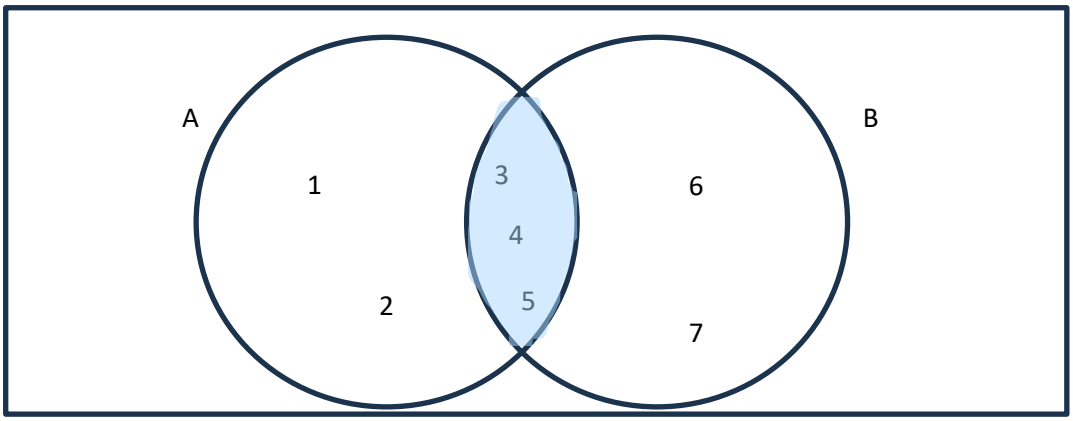
4. Question 1: Let A = {1, 2, 3} and B = {3, 4, 5}. Find the union of sets A and B. Draw Venn Diagram. **(2 Marks)**

$(A \cup B) = \{1,2,3,4,5\}$

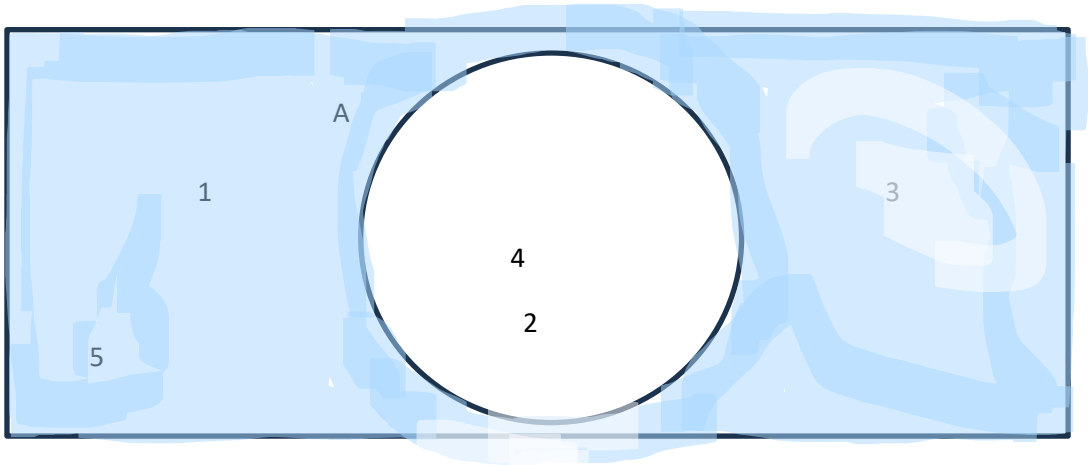


5. Question 2: If A = {1, 2, 3, 4, 5} and B = {3, 4, 5, 6, 7}, find the intersection of sets A and B. Draw Venn Diagram. **(2 Marks)**

   (A ∩ B)=3,4,5

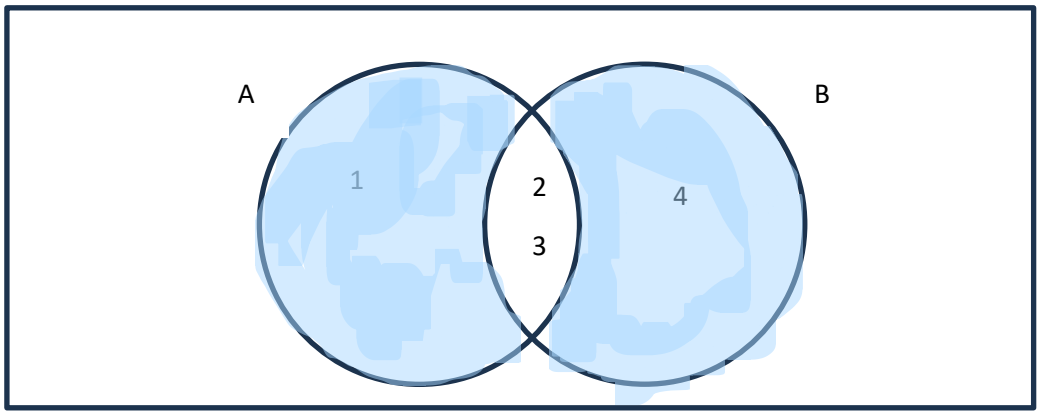


6. Question 3: Given the universal set U = {1, 2, 3, 4, 5}, find the complement of set A = {2, 4}. Draw Venn Diagram. **(2 Marks)**
   A'= {1,3,5}



7. Question 4: Let A = {1, 2, 3} and B = {2, 3, 4}. Find the set difference A - B. Draw Venn Diagram. **(2 Marks)**

   A-B={1,4}



TOTAL 20 Marks